

and

$$\begin{aligned}
 |\cosh \pi z| &= \left| \frac{e^{\pi z} + e^{-\pi z}}{2} \right| \\
 &\geq \frac{1}{2} \left| |e^{\pi z}| - |e^{-\pi z}| \right| \\
 &\geq \frac{1}{2} (e^{\pi R} - e^{-\pi R}) \\
 &\rightarrow \infty \quad \text{as } R \rightarrow \infty,
 \end{aligned}$$

which shows that the integral over the vertical segment on the right goes to 0 as $R \rightarrow \infty$. A similar argument shows that the integral of f over the vertical segment on the left also goes to 0 as $R \rightarrow \infty$. Finally, we see that if I denotes the integral we wish to calculate, then the integral of f over the top side of the rectangle (with the orientation from right to left) is simply $-e^{4\pi\xi}I$ where we have used the fact that $\cosh \pi\zeta$ is periodic with period $2i$. In the limit as R tends to infinity, the residue formula gives

$$\begin{aligned}
 I - e^{4\pi\xi}I &= 2\pi i \left(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i} \right) \\
 &= -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}),
 \end{aligned}$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$, we find that

$$I = 2 \frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = 2 \frac{e^{\pi\xi} - e^{-\pi\xi}}{(e^{\pi\xi} - e^{-\pi\xi})(e^{\pi\xi} + e^{-\pi\xi})} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}$$

as claimed.

A similar argument actually establishes the following formula:

$$\int_{-\infty}^{\infty} e^{-2\pi i x \xi} \frac{\sin \pi a}{\cosh \pi x + \cos \pi a} dx = \frac{2 \sinh 2\pi a \xi}{\sinh 2\pi \xi}$$

whenever $0 < a < 1$, and where $\sinh z = (e^z - e^{-z})/2$. We have proved above the particular case $a = 1/2$. This identity can be used to determine an explicit formula for the Poisson kernel for the strip (see Problem 3 in Chapter 5 of Book I), or to prove the sum of two squares theorem, as we shall see in Chapter 10.

3 Singularities and meromorphic functions

Returning to Section 1, we see that we have described the analytical character of a function near a pole. We now turn our attention to the other types of isolated singularities.

Let f be a function holomorphic in an open set Ω except possibly at one point z_0 in Ω . If we can define f at z_0 in such a way that f becomes holomorphic in all of Ω , we say that z_0 is a **removable** singularity for f .

Theorem 3.1 (Riemann's theorem on removable singularities)

Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω . If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.

Proof. Since the problem is local we may consider a small disc D centered at z_0 and whose closure is contained in Ω . Let C denote the boundary circle of that disc with the usual positive orientation. We shall prove that if $z \in D$ and $z \neq z_0$, then under the assumptions of the theorem we have

$$(4) \quad f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Since an application of Theorem 5.4 in the previous chapter proves that the right-hand side of equation (4) defines a holomorphic function on *all* of D that agrees with $f(z)$ when $z \neq z_0$, this gives us the desired extension.

To prove formula (4) we fix $z \in D$ with $z \neq z_0$ and use the familiar toy contour illustrated in Figure 4.

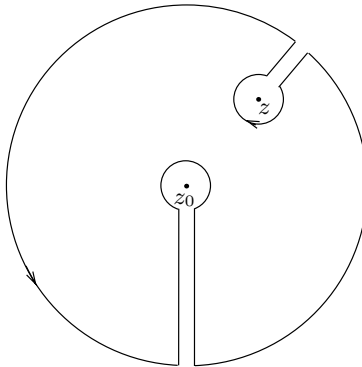


Figure 4. The multiple keyhole contour in the proof of Riemann's theorem

The multiple keyhole avoids the two points z and z_0 . Letting the sides of the corridors get closer to each other, and finally overlap, in the limit

we get a cancellation:

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z} d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z} d\zeta = 0,$$

where γ_ϵ and γ'_ϵ are small circles of radius ϵ with negative orientation and centered at z and z_0 respectively. Copying the argument used in the proof of the Cauchy integral formula in Section 4 of Chapter 2, we find that

$$\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z} d\zeta = -2\pi i f(z).$$

For the second integral, we use the assumption that f is bounded and that since ϵ is small, ζ stays away from z , and therefore

$$\left| \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z} d\zeta \right| \leq C\epsilon.$$

Letting ϵ tend to 0 proves our contention and concludes the proof of the extension formula (4).

Surprisingly, we may deduce from Riemann's theorem a characterization of poles in terms of the behavior of the function in a neighborhood of a singularity.

Corollary 3.2 *Suppose that f has an isolated singularity at the point z_0 . Then z_0 is a pole of f if and only if $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$.*

Proof. If z_0 is a pole, then we know that $1/f$ has a zero at z_0 , and therefore $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$. Conversely, suppose that this condition holds. Then $1/f$ is bounded near z_0 , and in fact $1/|f(z)| \rightarrow 0$ as $z \rightarrow z_0$. Therefore, $1/f$ has a removable singularity at z_0 and must vanish there. This proves the converse, namely that z_0 is a pole.

Isolated singularities belong to one of three categories:

- Removable singularities (f bounded near z_0)
- Pole singularities ($|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$)
- Essential singularities.

By default, any singularity that is not removable or a pole is defined to be an **essential singularity**. For example, the function $e^{1/z}$ discussed at the very beginning of Section 1 has an essential singularity at

0. We already observed the wild behavior of this function near the origin. Contrary to the controlled behavior of a holomorphic function near a removable singularity or a pole, it is typical for a holomorphic function to behave erratically near an essential singularity. The next theorem clarifies this.

Theorem 3.3 (Casorati-Weierstrass) *Suppose f is holomorphic in the punctured disc $D_r(z_0) - \{z_0\}$ and has an essential singularity at z_0 . Then, the image of $D_r(z_0) - \{z_0\}$ under f is dense in the complex plane.*

Proof. We argue by contradiction. Assume that the range of f is not dense, so that there exists $w \in \mathbb{C}$ and $\delta > 0$ such that

$$|f(z) - w| > \delta \quad \text{for all } z \in D_r(z_0) - \{z_0\}.$$

We may therefore define a new function on $D_r(z_0) - \{z_0\}$ by

$$g(z) = \frac{1}{f(z) - w},$$

which is holomorphic on the punctured disc and bounded by $1/\delta$. Hence g has a removable singularity at z_0 by Theorem 3.1. If $g(z_0) \neq 0$, then $f(z) - w$ is holomorphic at z_0 , which contradicts the assumption that z_0 is an essential singularity. In the case that $g(z_0) = 0$, then $f(z) - w$ has a pole at z_0 also contradicting the nature of the singularity at z_0 . The proof is complete.

In fact, Picard proved a much stronger result. He showed that under the hypothesis of the above theorem, the function f takes on every complex value infinitely many times with at most one exception. Although we shall not give a proof of this remarkable result, a simpler version of it will follow from our study of entire functions in a later chapter. See Exercise 11 in Chapter 5.

We now turn to functions with only isolated singularities that are poles. A function f on an open set Ω is **meromorphic** if there exists a sequence of points $\{z_0, z_1, z_2, \dots\}$ that has no limit points in Ω , and such that

- (i) the function f is holomorphic in $\Omega - \{z_0, z_1, z_2, \dots\}$, and
- (ii) f has poles at the points $\{z_0, z_1, z_2, \dots\}$.

It is also useful to discuss functions that are meromorphic in the extended complex plane. If a function is holomorphic for all large values of

z , we can describe its behavior at infinity using the tripartite distinction we have used to classify singularities at finite values of z . Thus, if f is holomorphic for all large values of z , we consider $F(z) = f(1/z)$, which is now holomorphic in a deleted neighborhood of the origin. We say that f has a **pole at infinity** if F has a pole at the origin. Similarly, we can speak of f having an **essential singularity at infinity**, or a **removable singularity** (hence holomorphic) **at infinity** in terms of the corresponding behavior of F at 0. A meromorphic function in the complex plane that is either holomorphic at infinity or has a pole at infinity is said to be **meromorphic in the extended complex plane**.

At this stage we return to the principle mentioned at the beginning of the chapter. Here we can see it in its simplest form.

Theorem 3.4 *The meromorphic functions in the extended complex plane are the rational functions.*

Proof. Suppose that f is meromorphic in the extended plane. Then $f(1/z)$ has either a pole or a removable singularity at 0, and in either case it must be holomorphic in a deleted neighborhood of the origin, so that the function f can have only finitely many poles in the plane, say at $z_1, \dots, z_n$. The idea is to subtract from f its principal parts at all its poles including the one at infinity. Near each pole $z_k \in \mathbb{C}$ we can write

$$f(z) = f_k(z) + g_k(z),$$

where $f_k(z)$ is the principal part of f at z_k and g_k is holomorphic in a (full) neighborhood of z_k . In particular, f_k is a polynomial in $1/(z - z_k)$. Similarly, we can write

$$f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z),$$

where $\tilde{g}_\infty$ is holomorphic in a neighborhood of the origin and $\tilde{f}_\infty$ is the principal part of $f(1/z)$ at 0, that is, a polynomial in $1/z$. Finally, let $f_\infty(z) = \tilde{f}_\infty(1/z)$.

We contend that the function $H = f - f_\infty - \sum_{k=1}^n f_k$ is entire and bounded. Indeed, near the pole z_k we subtracted the principal part of f so that the function H has a removable singularity there. Also, $H(1/z)$ is bounded for z near 0 since we subtracted the principal part of the pole at ∞ . This proves our contention, and by Liouville's theorem we conclude that H is constant. From the definition of H , we find that f is a rational function, as was to be shown.

Note that as a consequence, a rational function is determined up to a multiplicative constant by prescribing the locations and multiplicities of its zeros and poles.

The Riemann sphere

The extended complex plane, which consists of $\mathbb{C}$ and the point at infinity, has a convenient geometric interpretation, which we briefly discuss here.

Consider the Euclidean space $\mathbb{R}^3$ with coordinates (X, Y, Z) where the XY -plane is identified with $\mathbb{C}$. We denote by $\mathbb{S}$ the sphere centered at $(0, 0, 1/2)$ and of radius $1/2$; this sphere is of unit diameter and lies on top of the origin of the complex plane as pictured in Figure 5. Also, we let $\mathcal{N} = (0, 0, 1)$ denote the north pole of the sphere.

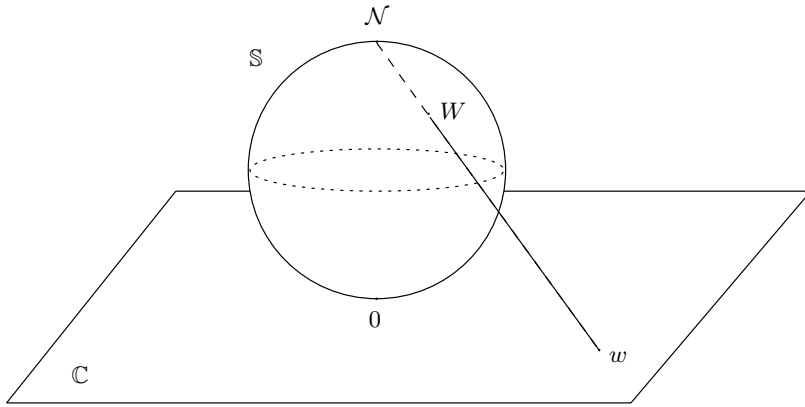


Figure 5. The Riemann sphere $\mathbb{S}$ and stereographic projection

Given any point $W = (X, Y, Z)$ on $\mathbb{S}$ different from the north pole, the line joining $\mathcal{N}$ and W intersects the XY -plane in a single point which we denote by $w = x + iy$; w is called the **stereographic projection** of W (see Figure 5). Conversely, given any point w in $\mathbb{C}$, the line joining $\mathcal{N}$ and $w = (x, y, 0)$ intersects the sphere at $\mathcal{N}$ and another point, which we call W . This geometric construction gives a bijective correspondence between points on the punctured sphere $\mathbb{S} - \{\mathcal{N}\}$ and the complex plane; it is described analytically by the formulas

$$x = \frac{X}{1 - Z} \quad \text{and} \quad y = \frac{Y}{1 - Z},$$

giving w in terms of W , and

$$X = \frac{x}{x^2 + y^2 + 1}, \quad Y = \frac{y}{x^2 + y^2 + 1}, \quad \text{and} \quad Z = \frac{x^2 + y^2}{x^2 + y^2 + 1}$$

giving W in terms of w . Intuitively, we have wrapped the complex plane onto the punctured sphere $\mathbb{S} - \{\mathcal{N}\}$.

As the point w goes to infinity in $\mathbb{C}$ (in the sense that $|w| \rightarrow \infty$) the corresponding point W on $\mathbb{S}$ comes arbitrarily close to $\mathcal{N}$. This simple observation makes $\mathcal{N}$ a natural candidate for the so-called “point at infinity.” Identifying infinity with the point $\mathcal{N}$ on $\mathbb{S}$, we see that the extended complex plane can be visualized as the full two-dimensional sphere $\mathbb{S}$; this is the **Riemann sphere**. Since this construction takes the unbounded set $\mathbb{C}$ into the compact set $\mathbb{S}$ by adding one point, the Riemann sphere is sometimes called the **one-point compactification** of $\mathbb{C}$.

An important consequence of this interpretation is the following: although the point at infinity required special attention when considered separately from $\mathbb{C}$, it now finds itself on equal footing with all other points on $\mathbb{S}$. In particular, a meromorphic function on the extended complex plane can be thought of as a map from $\mathbb{S}$ to itself, where the image of a pole is now a tractable point on $\mathbb{S}$, namely $\mathcal{N}$. For these reasons (and others) the Riemann sphere provides good geometrical insight into the structure of $\mathbb{C}$ as well as the theory of meromorphic functions.

4 The argument principle and applications

We anticipate our discussion of the logarithm (in Section 6) with a few comments. In general, the function $\log f(z)$ is “multiple-valued” because it cannot be defined unambiguously on the set where $f(z) \neq 0$. However it is to be defined, it must equal $\log |f(z)| + i \arg f(z)$, where $\log |f(z)|$ is the usual real-variable logarithm of the positive quantity $|f(z)|$ (and hence is defined unambiguously), while $\arg f(z)$ is some determination of the argument (up to an additive integral multiple of 2π). Note that in any case, the derivative of $\log f(z)$ is $f'(z)/f(z)$ which is single-valued, and the integral

$$\int_{\gamma} \frac{f'(z)}{f(z)} dz$$

can be interpreted as the change in the argument of f as z traverses the curve γ . Moreover, assuming the curve is closed, this change of argument is determined entirely by the zeros and poles of f inside γ . We now formulate this fact as a precise theorem.

We begin with the observation that while the additivity formula

$$\log(f_1 f_2) = \log f_1 + \log f_2$$

fails in general (as we shall see below), the additivity can be restored to the corresponding derivatives. This is confirmed by the following